

SECTION 9

FUNCTIONAL SIGNAL DESCRIPTIONS

This section contains a brief description of the input and output signals for the MC68881/MC68882 (FPCP) floating-point coprocessor. The signals are functionally organized into groups as shown in Figure 9-1.

NOTE

The terms **assertion** and **negation** are used extensively to avoid confusion when describing "active-low" and "active-high" signals. The term **assert** or **assertion** is used to indicate that a signal is active or true, regardless of whether that level is represented by a high or low voltage. The term **negate** or **negation** is used to indicate that a signal is inactive or false.

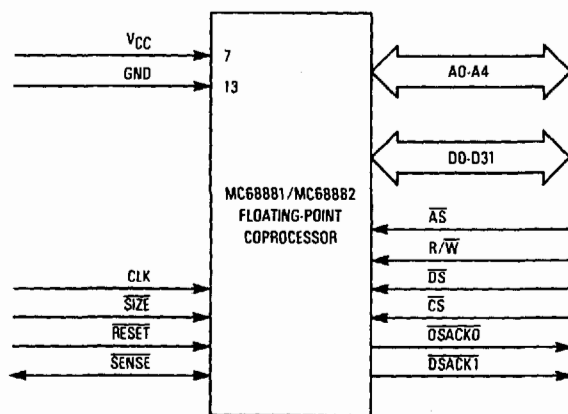


Figure 9-1. MC68881/MC68882 Input/Output Signals

9.1 ADDRESS BUS (A4-A0)

These active-high address line inputs are used by the main processor to select the coprocessor interface register locations located in the CPU address space. These lines control the register selection as listed in Table 9-1.

When the FPCP operates with an 8-bit data bus, the A0 pin is used as an address signal for byte accesses of the coprocessor interface registers. When the FPCP operates with a 16- or 32-bit system data bus, both the A0 and SIZE pins are strapped high and/or low as listed in Table 9-2.

Table 9-1. Coprocessor Interface Register Selection

A4-A0	Offset	Width	Type	Register
0000x	\$00	16	Read	Response
0001x	\$02	16	Write	Control
0010x	\$04	16	Read	Save
0011x	\$06	16	Read Write	Restore
0100x	\$08	16	—	(Reserved)
0101x	\$0A	16	Write	Command
0110x	\$0C	16	—	(Reserved)
0111x	\$0E	16	Write	Condition
100xx	\$10	32	Read/Write	Operand
1010x	\$14	16	Read	Register Select
1011x	\$16	16	—	(Reserved)
110xx	\$18	32	Write	Instruction Address
111xx*	\$1C	32	Read Write	Operand Address

*Not used by the MC68881 or MC68882

Table 9-2. System Data Bus Size Configuration

A0	Size	Data Bus
—	Low	8-Bit
Low	High	16-Bit
High	High	32-Bit

9.2 DATA BUS (D31–D0)

This 32-bit, bidirectional, three-state bus serves as the general-purpose data path between the MC68020/MC68030 (MPU) and the FPCP. Regardless of whether the FPCP is operating as a coprocessor or a peripheral processor, all interprocessor transfers of instruction information, operand data, status information, and requests for service occur as standard M68000 bus cycles.

The FPCP can operate with an 8-, 16-, or 32-bit system data bus. To operate with the required system data bus size, both the A0 and $\overline{\text{SIZE}}$ pins must be connected specifically for that applicable bus size. (Refer to 9.1 ADDRESS BUS (A4–A0) and 9.3 SIZE ($\overline{\text{SIZE}}$) for further details.)

9.3 SIZE ($\overline{\text{SIZE}}$)

This active-low input signal is used in conjunction with the A0 pin to configure the FPCP for operation over an 8-, 16-, or 32-bit system data bus. When the FPCP is configured to operate over a 16- or 32-bit system data bus, the $\overline{\text{SIZE}}$ and A0 pins must be strapped as listed in Table 9-2.

This active-low input signal indicates that there is a valid address on the address bus, and both the chip select ($\overline{CS}$) and read/write ($R/\overline{W}$) signal lines are valid.

9.5 CHIP SELECT ($\overline{CS}$)

This active-low input signal enables the main processor access to the FPCP coprocessor interface registers. When operating the FPCP as a peripheral processor, the chip-select decode is system dependent (i.e., like the chip select on any peripheral). The $\overline{CS}$ signal must be valid (either asserted or negated) when $\overline{AS}$ is asserted. Refer to **10.3 Chip Select Timing** for further discussion of timing restrictions for this signal.

9.6 READ/WRITE ($R/\overline{W}$)

This input signal indicates the direction of a bus transaction (read/write) by the main processor. A logic high (1) indicates a read from the FPCP, and a logic low (0) indicates a write to the FPCP. The $R/\overline{W}$ signal must be valid when $\overline{AS}$ is asserted.

9.7 DATA STROBE ($\overline{DS}$)

This active-low input signal indicates that there is valid data on the data bus during a write bus cycle.

9.8 DATA TRANSFER AND SIZE ACKNOWLEDGE ($\overline{DSACK1}$, $\overline{DSACK0}$)

These active-low, three-state output signals indicate the completion of a bus cycle to the main processor. The FPCP asserts both the $\overline{DSACK1}$ and $\overline{DSACK0}$ signals when the MPU asserts $\overline{CS}$.

If the bus cycle is a main processor read, the FPCP asserts $\overline{DSACK1}$ and $\overline{DSACK0}$ signals to indicate that the information on the data bus is valid. (Both $\overline{DSACK}$ signals can be asserted in advance of the valid data being placed on the bus.) If the bus cycle is a main processor write to the FPCP, $\overline{DSACK1}$ and $\overline{DSACK0}$ are used to acknowledge acceptance of the data by the FPCP.

The FPCP also uses $\overline{DSACK1}$ and $\overline{DSACK0}$ signals to dynamically indicate to the MPU the port size (system data bus width) on a cycle-by-cycle basis. Depending upon which of the two $\overline{DSACK}$ pins is asserted for a bus cycle, the MPU assumes data has been transferred to/from an 8-, 16-, or 32-bit wide data port. Table 9-3 lists the $\overline{DSACK}$ assertions that are used by the FPCP for the various bus cycles over the various system data bus configurations. Refer to **10.1 BASIC TRANSFER MECHANISM OVERVIEW** for details of data bus utilization by the FPCP.

Table 9-3 indicates that all accesses using a 32-bit bus with A4 equal to zero are to 16-bit registers. The FPCP implements all 16-bit coprocessor interface registers on data lines D31–D16 (to eliminate the need for on-chip multiplexors); however, the MPU expects 16-bit registers that are located in a 32-bit port at odd word addresses (A1 = 1) to be implemented on data lines D15–D0. For accesses to these registers when configured for 32-bit

Table 9-3. DSACK Assertions

Data Bus	A4	$\overline{\text{DSACK1}}$	$\overline{\text{DSACK0}}$	Comments
32-Bit	1	L	L	Valid Data on D31–D0
32-Bit	0	L	H	Valid Data on D31–D16
16-Bit	x	L	H	Valid Data on D31–D16 or D15–D0
8-Bit	x	H	L	Valid Data on D31–D24, D23–D16, D15–D8, or D7–D0
All	x	H	H	Insert Wait States in Current Bus Cycle

Table 9-3 indicates that all accesses using a 32-bit bus with A4 equal to zero are to 16-bit registers. The FPCP implements all 16-bit coprocessor interface registers on data lines D31–D16 (to eliminate the need for on-chip multiplexors); however, the MPU expects 16-bit registers that are located in a 32-bit port at odd word addresses (A1 = 1) to be implemented on data lines D15–D0. For accesses to these registers when configured for 32-bit bus operation, the MC68881/M68882 generates DSACK signals as listed in Table 9-3 to indicate to the MPU that the valid data is on D31–D16 instead of on D15–D0.

External holding resistors are required to maintain both $\overline{\text{DSACK1}}$ and $\overline{\text{DSACK0}}$ high between bus cycles. In order to reduce the signal rise time, the $\overline{\text{DSACK1}}$ and $\overline{\text{DSACK0}}$ lines are actively pulled up (negated) by the FPCP following the rising edge of $\overline{\text{AS}}$ or $\overline{\text{DS}}$, and both DSACK lines are then three-stated (placed in the high-impedance state) to avoid interference with the next bus cycle.

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9.9 RESET ($\overline{\text{RESET}}$)

This active-low input signal causes the FPCP to initialize the floating-point data registers to nonsignaling not-a-numbers (NaNs) and clears the floating-point control, status, and instruction address registers.

When performing a powerup reset, external circuitry should keep the $\overline{\text{RESET}}$ line asserted for a minimum of four clock cycles after V_{CC} is within tolerance. This assures correct initialization of the FPCP when power is applied. For compatibility with all M68000 Family devices, 100 milliseconds should be used as the minimum.

When performing a reset of the FPCP after V_{CC} has been within tolerance for more than the initial powerup time, the $\overline{\text{RESET}}$ line must have an asserted pulse width greater than two clock cycles. For compatibility with all M68000 Family devices, 10 clock cycles should be used as the minimum.

9.10 CLOCK (CLK)

The FPCP clock input is a TTL-compatible signal that is internally buffered for development of the internal clock signals. The clock input must be a constant frequency square wave with no stretching or shaping techniques required. The clock should not be gated off at any time and must conform to minimum and maximum period and pulse width times.

This output pin may be used optionally as an additional GND pin or as an indicator to external hardware that the FPCP is present in the system. This signal is internally connected to the GND of the die, but it is not necessary to connect it to the external ground for correct device operation.

Figure 9-2 shows an example of a circuit to sense the presence of an FPCP in a socket prepared for it. The circuit asserts $\overline{\text{BERR}}$ when the MPU selects the coprocessor and no coprocessor is plugged in.

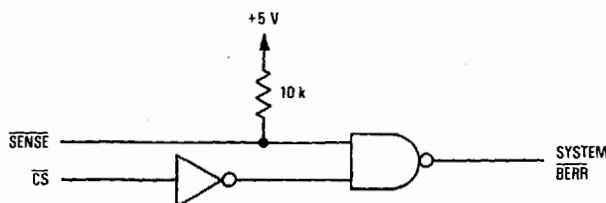


Figure 9-2. Sense Device Circuit Example

9.12 POWER (VCC and GND)

These pins provide the supply voltage and system reference level for the internal circuitry of the FPCP. Care should be taken to reduce the noise level on these pins with appropriate capacitive decoupling.

The FPCP is fabricated in Freescale's advanced HCMOS process and is capable of operating at clock speeds of 25 MHz. Although the use of CMOS for a device containing such a large number of transistors allows significantly reduced power consumption in comparison to an equivalent NMOS device, the high clock speed makes the characteristics of the power supplied to the part quite important. The power supply must be as free from noise as possible, and it must be able to supply large amounts of instantaneous current when the FPCP performs certain operations. In order to meet these requirements, more detailed attention should be given to the power supply connection to the FPCP than is required for older NMOS devices that operate at slower clock rates.

In order to provide a solid power supply interface, four VCC pins, eight primary GND pins, and two secondary GND pins are provided. This allows two VCC and GND pins to supply the power for the data bus, while the remaining VCC and GND pins are used by the internal logic and DSACK drivers. The two secondary GND pins are not intended to provide the main power supply interface, but merely to augment it as required. (One of these pins is the SENSE pin which may be used as an optional GND connection.)

Three VCC and four GND pin positions are reserved for future use by Freescale and should be connected appropriately in order to maintain pin compatibility with all future versions of the FPCP. Table 9-4 lists the VCC and GND pin assignments.

In order to reduce the amount of noise in the power supplied to the FPCP, common capacitive decoupling techniques should be observed. While there is no recommended

Table 9-4. VCC and GND Pin Assignments

Devices Supplied	VCC	GND
D31-D16	H8	J8
D15-D00	B8	B7
Internal Logic, DSACK1, DSACK0	E2, E9	A2, B2, B3, B4*, C3, E10, K3
Separate	—	C1
Extra	A1, B1, J2	A10, D2, F2, H9

*B4 is the SENSE pin and may be used optionally as a ground pin or to detect the presence of the FPCP in the system.

layout for this capacitive decoupling, it is suggested that a combination of low, middle, and high frequency filter capacitors be placed as close to the chip as possible. (For example, a set of 10 μ F, 0.1 μ F, and 330 pF capacitors in parallel provides filtering for nearly the entire frequency spectrum present in a digital system.) In a system that utilizes the MC68020 as the main processor, these capacitive decoupling practices should also be observed for the main processor. In particular, the 10 μ F “tank” capacitor should be reasonably close to both devices (since the two devices are typically placed next to each other on a board) to provide for the high instantaneous current requirements of both the MPU and the FPCP.

In addition to the capacitive decoupling of the power supply, care should be taken to ensure a low resistance connection between the FPCP VCC and GND pins and the system power supply traces. In particular, the connections to pins B7 and J8 (the GND pins for the data bus pins) must have very low resistance. This is necessary because a read of the FPCP can cause the data bus drivers to sink very large amounts of current to ground in order to pull the data bus signals low (if the data pattern that is read contains mostly zeros). If low resistance connections are not provided on pins B7 and J8, the ground potential internal to the package may rise, the fall time of the data signals may be increased, and the low output voltage noise margin may be reduced.

9.13 NO CONNECT (NC)

One pin of the FPCP package is designated as a no connect (NC). This pin position is reserved for future use by Freescale and should neither be used for signal routing nor connected to VCC or GND.

9.14 SIGNAL SUMMARY

Table 9-5 provides a summary of all the FPCP signals described in this section.

Table 9-5. Signal Summary

Signal Name	Mnemonic	Input/Output	Active State	Three State
Address Bus	A4–A0	Input	High	—
Data Bus	D31–D0	Input/Output	High	Yes
Size	SIZE	Input	Low	—
Address Strobe	$\overline{AS}$	Input	Low	—
Chip Select	$\overline{CS}$	Input	Low	—
Read/Write	R/W	Input	High/Low	—
Data Strobe	$\overline{DS}$	Input	Low	—
Data Transfer and Size Acknowledge	DSACK1, DSACK0	Output	Low	Yes
Reset	$\overline{RESET}$	Input	Low	—
Clock	CLK	Input	—	—
Sense Device	$\overline{SENSE}$	Input/Output	Low	No
Power Input	V _{CC}	Input	—	—
Ground	GND	Input	—	—

